COMPLETE ENGINEERING DESIGN PACKAGE

Tubular Daylight Collector Dome

Detailed prototype geometry, optical collection principles, mold data, component interfaces, fabrication sequence, and validation plan for a 14-inch nominal tubular daylight device.

DOCUMENT	DESIGN BASIS	STATUS	UNITS
Engineering design description	14-inch / 356 mm clear aperture	Prototype - Rev A	mm unless noted

Design intent

The green collector at the top of the daylight tube is treated as a clear, UV-stabilized optical dome with a directional prismatic sector and a reflective transition. The dome protects the opening, admits diffuse sky light, accepts high-angle direct sunlight, and redirects selected low-angle sunlight toward the reflective tube.

PARAMETER	DESIGN VALUE	PURPOSE
Clear tube aperture	Dia. 356	Matches 14-inch nominal tube
Dome outer base	Dia. 420	Optical entrance and sealing land
Dome rise	230	Broad sky view and useful incidence angles
Mounting flange	Dia. 460	Clamp, gasket, and fastener interface
Nominal sheet	5.0 optical PC	Allows finished wall >= 3.0 after forming
CPC transition	Dia. 404 to 356 x H 90	Guides light into the tube

ENGINEERING LIMIT
This is a preliminary prototype design, not a certified roof product. Wind, snow, hail, impact, fire, flashing, leakage, structural attachment, and local code compliance must be checked by qualified specialists before outdoor installation.

1. Geometry and dimensional definition

The collector is an ellipsoid of revolution rather than a perfect hemisphere. A slightly taller profile improves sky exposure and gives room for the directional prism sector while keeping a broad flange for sealing.

OUTER SURFACE EQUATION

Revolve the profile 360 degrees about the z-axis: $\mathbf{(r / 210)^2 + (z / 230)^2 = 1}$, for 0 <= z <= 230. Here r is radial distance from the axis and z is height above the dome base plane.

Male mold coordinates

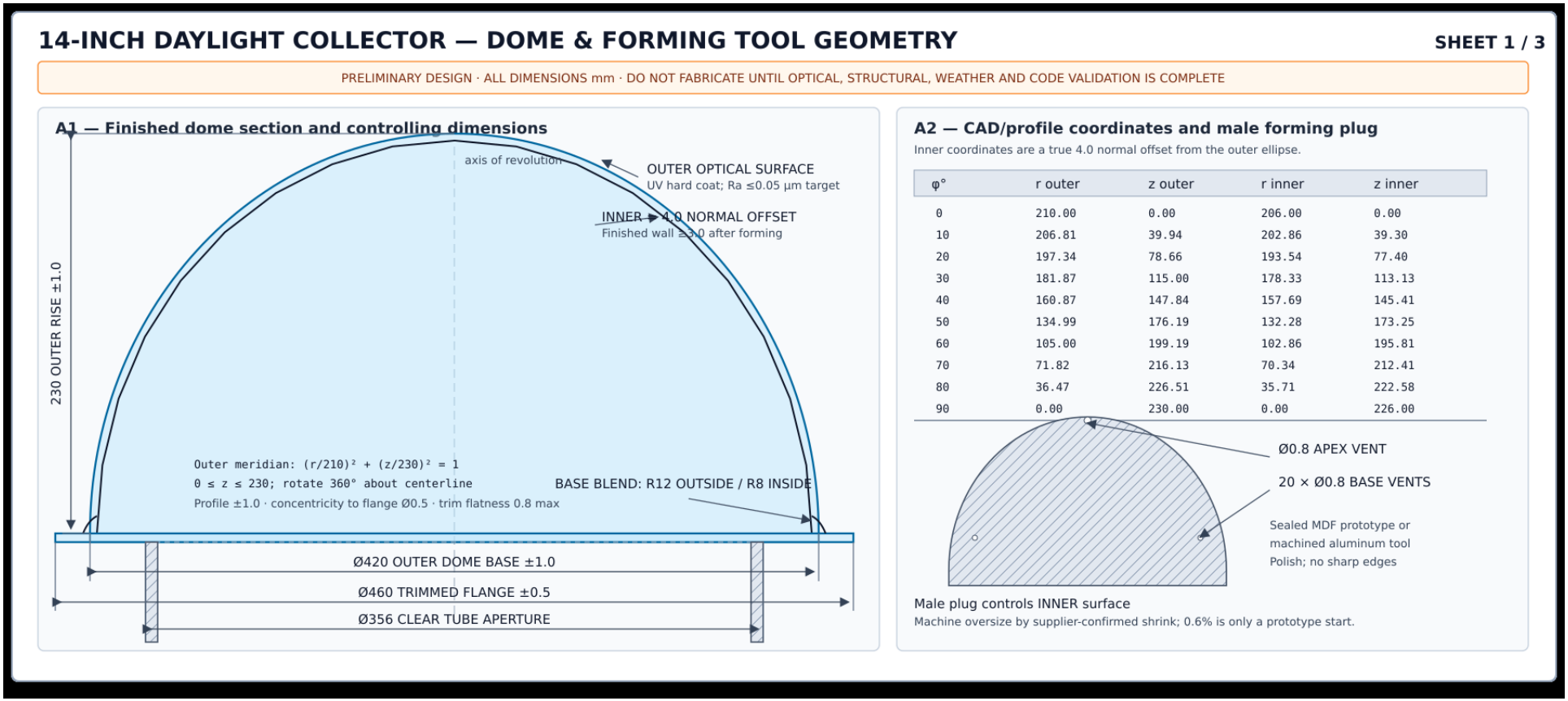
The inner surface is defined as a 4.0 mm normal offset from the outer optical surface. The male plug follows this inner surface. Use the values below to generate a revolved CAD spline; do not simply subtract 4.0 mm in the radial direction.

MERIDIAN ANGLE	OUTER r	OUTER z	INNER r	INNER z
0 deg	210.00	0.00	206.00	0.00
10 deg	206.81	39.94	202.86	39.30
20 deg	197.34	78.66	193.54	77.40
30 deg	181.87	115.00	178.33	113.13
40 deg	160.87	147.84	157.69	145.41
50 deg	134.99	176.19	132.28	173.25
60 deg	105.00	199.19	102.86	195.81
70 deg	71.82	216.13	70.34	212.41
80 deg	36.47	226.51	35.71	222.58
90 deg	0.00	230.00	0.00	226.00

Tool and finished-part tolerances

- Profile tolerance: +/-0.50 mm over the optical surface; no local waviness greater than 0.25 mm over a 50 mm chord.
- Base cut plane: flat within 0.50 mm; flange outer diameter Dia. 460 +/-0.50 mm.
- Finished wall: target 4.0 mm, minimum 3.0 mm at the apex and transition regions.
- Optical finish: polished tool surface; remove machining marks before forming. Do not flame-polish polycarbonate.
- Prototype shrink allowance: begin with 0.6 percent only as a trial value; confirm the actual resin/forming process and compensate the tool after measurement.

Overall dome geometry, mold section, and mounting interface



Sheet 1 establishes the ellipsoidal profile, clear aperture, flange, 4 mm normal wall definition, hole pattern, gasket, and clamp stack.

2. Optical collection design

The collector combines three optical functions: a smooth transparent dome for broad-angle transmission, a directional prism sector for low-angle sunlight, and a reflective compound parabolic concentrator (CPC) transition that guides admitted rays into the tube.

ELEMENT	STARTING DESIGN	OPTICAL ROLE
Clear dome	UV-stabilized optical polycarbonate; exterior UV and hard coat	Transmit diffuse sky light and direct sunlight over a wide incidence range
Directional prism sector	120-degree azimuth sector; meridian band 5 to 55 degrees	Refract low-altitude sunlight downward instead of allowing it to cross the dome and escape
Prism geometry	2.50 pitch; 1.17 depth; 35 / 55 degree facets; 1.0 base film	Starting geometry for ray tracing and prototype trials
CPC transition	Dia. 404 top, Dia. 356 bottom, height 90; protected silver reflectance target >= 98 percent	Capture edge rays and couple them to the reflective tube

Orientation

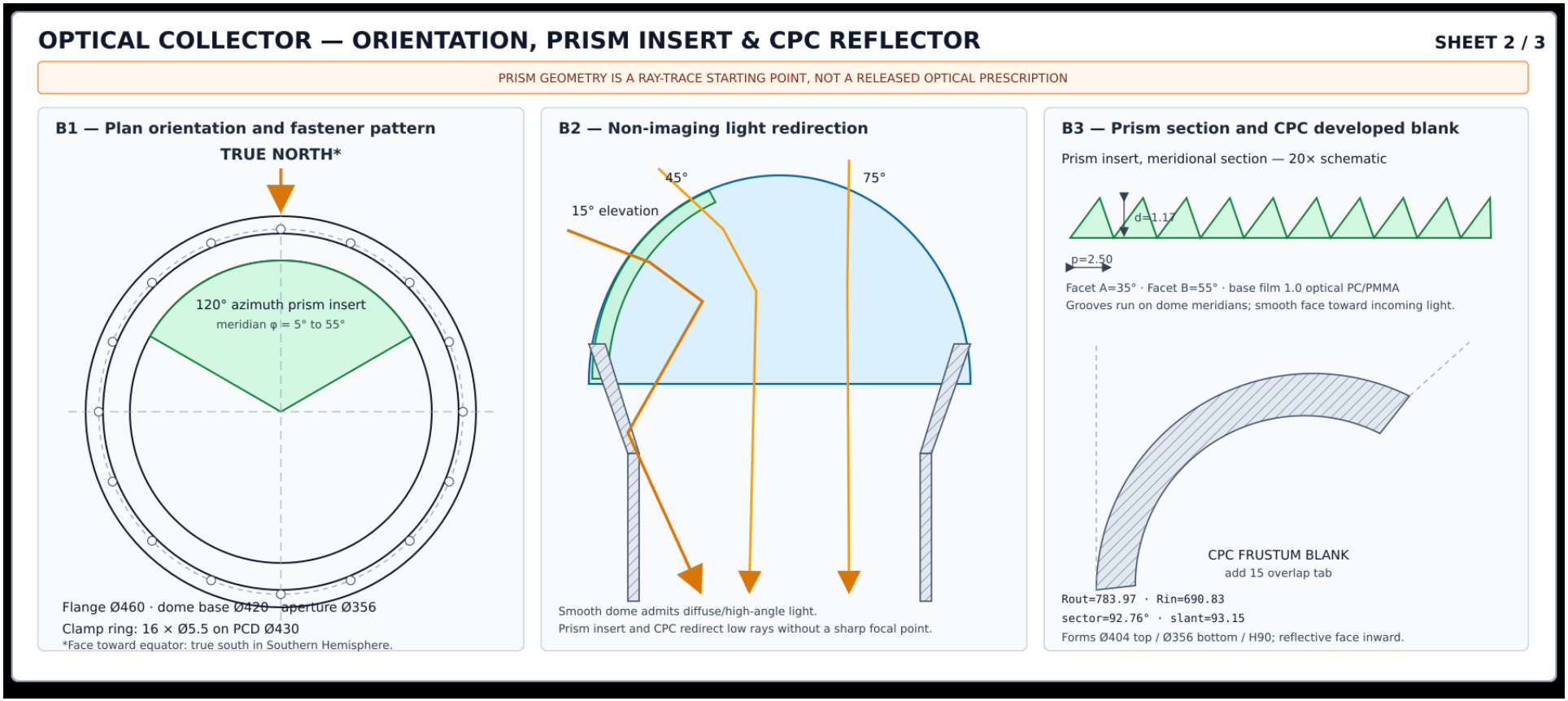
Place the prismatic sector toward the equator: face it generally south in the Northern Hemisphere and north in the Southern Hemisphere. Mark the equator-facing direction permanently on the clamp ring. Final azimuth should be chosen using the site latitude, roof pitch, surrounding shading, and the hours when useful daylight is most valuable.

Optical principles

- Snell refraction at the dome surface changes ray direction according to the refractive indices of air and polycarbonate. Use the actual supplier dispersion data in the ray model.
- Smooth, gradual curvature reduces sharp incidence transitions and supports daylight acceptance from a large portion of the sky vault.
- The asymmetric prism facets provide a preferred bend direction. The 35 / 55 degree facet pair is a manufacturable first trial, not a final prescription.
- The reflective transition must avoid wrinkles, oil canning, exposed adhesive, or open seams because each defect scatters light and reduces throughput.
- Evaluate annual useful lumens at the diffuser, not only noon peak illuminance. A wider, more uniform acceptance band is usually preferable to a narrow peak.

RAY-TRACE REQUIRED
Optimize the prism pitch, facet angles, sector width, and meridian coverage using the final resin index, wall thickness, latitude, weather data, tube reflectance, diffuser distribution, and roof orientation. The dimensions shown are a controlled prototype starting point.

Optical ray concepts, prism insert, CPC transition, and developed blank



Sheet 2 shows the directional daylight strategy and starting dimensions for the prismatic sector and reflective transition.

3. Components and interfaces

Use optical-grade, outdoor-qualified materials and keep all sealants, tapes, cleaners, and coatings chemically compatible with the selected polycarbonate grade. Obtain written compatibility statements from suppliers.

ITEM	COMPONENT	MATERIAL / GRADE	QTY	SIZE	NOTES
1	Collector dome	UV optical polycarbonate	1	5.0 sheet	Exterior UV and hard coat; clear; finished wall >= 3.0
2	Prism insert	Optical PC or PMMA	1	1.0 base	Embossed or molded; bond only with verified optical adhesive or mechanical retention
3	Clamp ring	5052-H32 or 6061-T6 Al	1	3.0	Anodized or powder coated; flatness <= 0.50
4	Gasket	UV and ozone resistant EPDM	1	3.0	Closed cell; design for 20 to 25 percent compression
5	CPC reflector	Aluminum + protected silver film	1	0.8	Reflectance target >= 98 percent; smooth, wrinkle-free optical surface
6	Fasteners	A4 stainless steel	16	M5 x 20	Bonded washer plus compression stop sleeve; avoid point loading
7	Tube collar	Aluminum or galvanized steel	1	0.8	Dia. 356 ID x 100 engagement; seal to tube system

Fastener and seal design

FEATURE	DOME	CLAMP RING	DESIGN CONTROL
Hole diameter	Dia. 7.0	Dia. 5.5	Allows dome thermal movement around M5 fastener
Bolt circle	PCD Dia. 430	PCD Dia. 430	16 positions, equally spaced at 22.5 degrees
Compression sleeve	7.3 to 7.6 long	Through ring/gasket stack	Stops over-compression and protects the polymer flange
Initial torque	Not torque-bearing	1.5 to 2.0 N m	Trial value; verify after leak and thermal cycling
Gasket compression	Uniform flange support	Continuous clamp	Target 20 to 25 percent; no local hard spots

THERMAL MOVEMENT
Do not rigidly lock the polycarbonate flange between metal parts. Oversize polymer holes, use bonded washers and stop sleeves, and allow the flange to expand without stress concentration.

ENGINEERING DRAWING - SHEET 3 OF 3

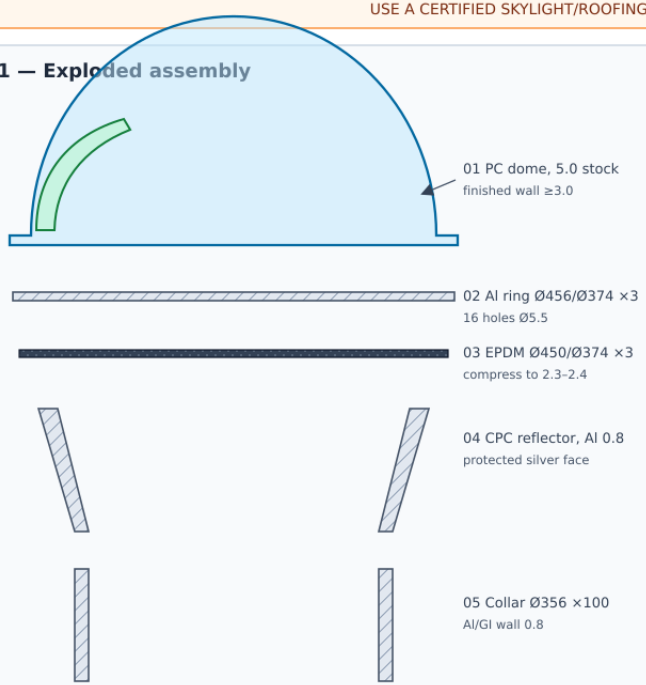
Small-batch fabrication sequence, assembly stack, and inspection controls

PARTS, SEALING DETAIL & PROTOTYPE FABRICATION SEQUENCE

SHEET 3 / 3

USE A CERTIFIED SKYLIGHT/ROOFING ENGINEER FOR WIND, SNOW, IMPACT, FIRE, FLASHING AND WATER-PENETRATION REQUIREMENTS

C1 — Exploded assembly

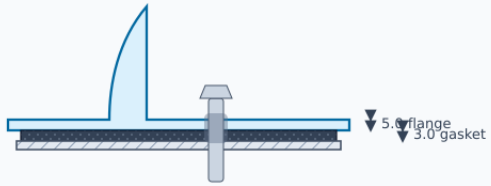


06 · 16× M5×20 A4-70 screws, bonded washers, stop sleeves and locknuts

C2 — Prototype bill of materials

Item	Qty	Material / grade	Finish	Critical requirement
01 Dome	1	UV optical PC sheet, 5.0	Exterior UV/hard coat	Clear; t min 3.0
02 Prism	1	Optical PC/PMMA, 1.0 base	Embossed prisms	Orientation
03 Ring	1	5052-H32 or 6061-T6 Al, 3.0	Anodize/powder coat	Flatness ≤0.5
04 Gasket	1	UV/ozone EPDM, 3.0	Closed cell	20–25% compression
05 CPC	1	Al sheet, 0.8	Protected silver ≥98% R	No wrinkles
06 Screws	16	M5×20 A4 stainless	Washer + stop sleeve	No point loading

C3 — Weather seal and thermal-expansion detail



Dome hole Ø7.0; clamp-ring hole Ø5.5
EPDM sleeve/washer centers M5 screw
Stop sleeve 7.3–7.6 controls gasket compression
Initial torque 1.5–2.0 N·m; verify after thermal cycle

C4 — Small-batch fabrication sequence

1 · Machine/polish male plug from profile table
2 · Predry PC per grade; start 120°C, 2–3 h
3 · Pressure/vacuum form in supplier window
4 · Cool on tool; trim; drill oversize holes
5 · Fit prism, CPC, gasket and clamp ring
6 · Leak, impact, load, thermal and ray tests

Never flame-polish PC; avoid incompatible solvent adhesives.

DESIGN BASIS: nominal 356 mm tube; northern-hemisphere orientation shown. Match materials and processing to the selected grade, latitude, roof pitch, climate loads and applicable skylight standards.

Sheet 3 is the workshop reference for plug tooling, thermoforming, trimming, prism and CPC installation, sealing, and prototype checks.

4. Prototype fabrication and assembly

The process below is intended for small-batch development. Confirm all temperature limits with the exact sheet grade and forming equipment; supplier values override generic starting conditions.

STEP	OPERATION	DETAIL / CONTROL
1	Build male plug	Machine sealed MDF for initial trials or aluminum for repeat production. Use the inner profile coordinate table and include a stable flange land.
2	Finish the tool	Seal and polish the optical surface. Provide one Dia. 0.8 apex vent and twenty Dia. 0.8 base vents, deburred and connected to vacuum.
3	Condition sheet	Keep protective masking in place where allowed. Predry the selected PC grade per supplier instructions; 120 C for 2 to 3 hours is only a starting trial.
4	Thermoform	Pressure form or vacuum form in the supplier process window. Heat uniformly, avoid chilling at the clamp, and record sheet temperature and draw ratio.
5	Cool on tool	Hold until the part is dimensionally stable. Premature release can cause springback, warpage, and flange distortion.
6	Trim and drill	Trim flange to Dia. 460. Drill Dia. 7.0 polymer holes on PCD Dia. 430 using sharp tools and backed support. Deburr without solvent.
7	Install optics	Fit the directional prism sector, then form and install the CPC. Keep optical surfaces clean and avoid trapped dust or adhesive squeeze-out.
8	Seal and clamp	Place EPDM gasket, clamp ring, stop sleeves, bonded washers, and M5 fasteners. Tighten gradually in a star pattern to 1.5 to 2.0 N m.

CPC developed blank

For the Dia. 404 to Dia. 356 by 90 mm conical transition, use a slant height of 93.15 mm. The starting annular-sector blank is: outer radius 783.97 mm, inner radius 690.83 mm, sector angle 92.76 degrees, plus a 15 mm overlap tab. Cut a paper or thin plastic trial first and verify the assembled diameters before cutting reflective material.

Assembly inspection

- Confirm equator-facing prism orientation and mark it on the clamp ring.
- Check the gasket is continuous, unwrinkled, and compressed evenly around the circumference.
- Confirm no fastener head or washer applies point load directly to the polycarbonate.
- Verify CPC seams are closed, the reflective face is undamaged, and the lower opening remains concentric with the tube.
- Use only lint-free cleaning materials and a cleaner approved by the sheet manufacturer.

5. Prototype validation plan

Treat the first articles as instrumented prototypes. Record material batch, sheet thickness map, tool temperature, forming temperatures, cooling time, final dimensions, and all test outcomes so the design can be corrected before release.

TEST	METHOD	INITIAL ACCEPTANCE TARGET
Dimensional inspection	CMM, template, or radial height gauge at the tabulated meridians	Profile within +/-0.50; flange flatness <= 0.50; minimum wall >= 3.0
Water leakage	Spray/rain test with installed gasket and representative roof interface	No water beyond the designed drainage/seal plane
Thermal cycle	Cycle representative hot-sun and cold-night temperatures; inspect at least 10 cycles	No cracks, crazing, fastener distress, seal displacement, or loss of torque control
Impact / hail screening	Use the applicable local skylight impact method and safety factors	No penetration or dangerous fragmentation; certify through a qualified laboratory
Optical throughput	Measure outdoor illuminance or luminous flux above the dome and at the diffuser	Stable transmission; compare smooth dome, prism prototype, and production-intent assembly
Directional response	Rotate a calibrated collimated source or test outdoors across solar altitude and azimuth	Prism sector increases useful tube-coupled light in the intended low-sun range without unacceptable glare
Environmental durability	Accelerated UV, humidity, adhesion, and coating checks using supplier and local standards	No unacceptable yellowing, haze, delamination, or reflective-film corrosion

Release gates

- Freeze the dome tool only after at least three measured parts show repeatable profile and wall distribution.
- Freeze prism geometry only after ray tracing is correlated with physical directional-light tests.
- Freeze clamp torque and stop-sleeve length only after leakage and thermal-cycle tests.
- Do not sell or permanently roof-mount the unit until a qualified engineer and test laboratory confirm the applicable structural, weather, fire, and safety requirements.

References and technical basis

- VELUX Commercial. Tubular daylighting sizing guidance. https://commercial.veluxusa.com/products/tubular/commercial-suntunnel/sizing
- POLYVANTIS. LEXAN solid polycarbonate sheet handling and processing resources. https://www.polyvantis.com/en/products/handling-care/polycarbonate/lexan-sheet/lexan-solid-sheet
- Covestro. Makrolon polycarbonate product information. https://solutions.covestro.com/en/brands/makrolon
- US Patent 5,655,339. Prismatic tubular skylight principle. https://patents.google.com/patent/US5655339A/en
- EP Patent 1,931,837 B1. Variable prism daylight-collection principle. https://patents.google.com/patent/EP1931837B1/en
- VELUX Commercial. Tubular daylighting technical resources. https://commercial.veluxusa.com/resources/tubular

FINAL RESPONSIBILITY

A roof-mounted daylight collector is a building-envelope and overhead safety component. The fabricator and installer remain responsible for material traceability, optical performance, structural design, drainage, flashing, sealing, fall protection, maintenance access, and all applicable approvals.